

1 Assignment Problems Part 1

1.1 Problem 1: K-Means Clustering and Rand Index

Consider the following data points in a 2D space:

$$P_1 = (1, 2), P_2 = (1.5, 1.8), P_3 = (5, 8), P_4 = (8, 8), P_5 = (1, 0.6), P_6 = (9, 11)$$

Assume initial cluster centers are $C_1 = (1, 1)$ and $C_2 = (8, 9)$.

1. Perform two rounds of the K-Means algorithm ($k = 2$) to update the cluster assignments and the cluster centers. Show your calculations for each step. (Hint: Use Euclidean distance to assign points to the nearest cluster center. After each round, recalculate the cluster centers as the mean of the points assigned to each cluster.)
2. After the two rounds, you will have two cluster assignments. Assume the "true" clustering of the data points is: Cluster A = $\{P_1, P_2, P_5\}$, Cluster B = $\{P_3, P_4, P_6\}$. Calculate the Rand Index (RI) between the clustering obtained after two rounds of K-Means and this true clustering. Show your calculations.

1.2 Problem 2: Naive Bayes Classification (Modified Example)

A small online bookstore wants to predict whether a customer will 'Like Books' (Yes/No) based on their 'Age Group' (Young, Adult, Senior) and 'Spends on Hobbies' (Low, High).

1.2.1 Training Data

Age Group	Spends on Hobbies	Likes Books
Young	Low	No
Adult	High	Yes
Adult	High	Yes
Senior	Low	Yes
Senior	High	Yes
Senior	Low	No
Young	Low	No
Young	High	Yes
Adult	High	Yes
Young	High	Yes
Senior	High	No
Young	Low	Yes
Adult	High	Yes
Adult	Low	Yes

Table 1: Book Liking Data

Using Naive Bayes classification, predict whether a customer with 'Age Group' = Young and 'Spends on Hobbies' = High will 'Like Books' (Yes or No). Show your calculations for prior probabilities and likelihood probabilities.

1.3 Problem 3: K-Nearest Neighbors (KNN) Classification

Consider a dataset for classifying fruits based on 'Weight (in grams)' and 'Size (1-small, 5-large)'. We want to predict the 'Fruit Type' (Apple or Banana) for a new fruit with Weight = 180 grams and Size = 4, using KNN with $k = 3$.

1.3.1 Training Data

Calculate the Euclidean distance for each training data point to the new fruit and determine the 3 nearest neighbors. Based on the majority class among these neighbors, predict the 'Fruit Type'.

Weight	Size	Fruit Type
150	3	Apple
170	4	Apple
120	2	Banana
190	4	Apple
130	3	Banana
160	3	Apple
110	1	Banana
185	5	Apple

Table 2: Fruit Classification Data

1.4 Problem 4: Decision Tree - Gini Index and Information Gain for Split Evaluation

Consider a dataset with a binary target variable 'Play Game' (Yes/No) and one feature 'Weather' (Sunny, Cloudy, Rainy). We want to decide the best way to split the data based on 'Weather'.

1.4.1 Dataset

Weather	Play Game
Sunny	Yes
Cloudy	Yes
Rainy	No
Sunny	Yes
Cloudy	No
Rainy	No
Sunny	No
Cloudy	Yes
Rainy	No
Sunny	Yes

Table 3: Weather and Game Playing Data

Calculate the Gini Index of the initial dataset. Then, consider splitting the data based on each unique value of 'Weather' (Sunny, Cloudy, Rainy) and calculate the Gini Index of the split for each case. Also, calculate the Information Gain for splitting on 'Weather' using Entropy. (Hint: Entropy of a dataset S is $Entropy(S) = -\sum_i p_i \log_2(p_i)$, where p_i is the proportion of each class. Information Gain $IG(S, A) = Entropy(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$). Based on your calculations, which split appears better according to Gini Index and Information Gain?

1.5 Problem 5: Linear Regression

A small company wants to understand the relationship between 'Advertising Spend (in \$1000s)' and 'Sales (in \$1000s)'. They have the following data for the last 5 months:

Advertising Spend	Sales
2	5
3	7
4	9
5	11
6	13

Table 4: Advertising and Sales Data

Assuming a simple linear regression model ($Sales = \beta_0 + \beta_1 \times AdvertisingSpend$), calculate the values of the slope (β_1) and the y-intercept (β_0) using the formulas:

$$\beta_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$
$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

where x_i is the advertising spend for month i , y_i is the sales for month i , $\bar{x}$ is the mean advertising spend, and $\bar{y}$ is the mean sales.

1.6 Problem 6: Decision Tree - Best First Split Attribute

Consider the following dataset for predicting 'Loan Approval' (Yes/No) based on 'Credit Score' (High, Low) and 'Income' (High, Low).

1.6.1 Dataset

Credit Score	Income	Loan Approval
High	High	Yes
Low	Low	No
High	Low	Yes
Low	High	Yes
High	High	Yes
Low	Low	No
High	High	Yes
Low	High	No

Table 5: Loan Approval Data

Calculate the Information Gain and Gini Index for Gain for splitting the data on the 'Credit Score' attribute and the 'Income' attribute. Based on these

calculations, which attribute would be the best choice for the first split in a decision tree? (Hint: Gini Gain is the reduction in Gini Index after the split: $GiniGain(S, A) = Gini(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Gini(S_v)$).

1.7 Problem 7: Confusion Matrix and Classification Metrics

A company conducted a survey of 5000 customers to predict whether they would 'Renew Subscription' (Yes/No). The actual renewal status after one year and the prediction from their model are summarized below.

Out of 4000 customers who did not actually renew, the model correctly predicted 'No' for 3800 and incorrectly predicted 'Yes' for 200. Out of 1000 customers who actually renewed, the model correctly predicted 'Yes' for 700 and incorrectly predicted 'No' for 300.

1. Construct a confusion matrix for the prediction results. Clearly label the axes and the values in each cell (True Positive, True Negative, False Positive, False Negative).
2. Based on the confusion matrix, compute the following classification metrics for the 'Renew Subscription' = 'Yes' class:
 - Precision
 - Specificity
 - Recall (Sensitivity)
 - Accuracy
 - F1-Score

Show your calculations for each metric.

2 Evaluation Metrics from Confusion Matrix

Consider a binary classification problem where a model predicts whether an email is "Spam" (Positive) or "Not Spam" (Negative). The confusion matrix summarizes the results as follows:

	Predicted: Spam (Positive)	Predicted: Not Spam (Negative)
Actual: Spam (Positive)	True Positives (TP)	False Negatives (FN)
Actual: Not Spam (Negative)	False Positives (FP)	True Negatives (TN)

The following evaluation metrics can be defined based on the confusion matrix:

Precision

Precision measures the proportion of emails predicted as Spam that were actually Spam.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

Specificity

Specificity measures the proportion of emails that were actually Not Spam that were correctly predicted as Not Spam.

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

Recall (Sensitivity or True Positive Rate)

Recall measures the proportion of emails that were actually Spam that were correctly identified as Spam.

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Accuracy

Accuracy measures the overall proportion of emails that were correctly classified.

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{\text{True Positives (TP)} + \text{False Positives (FP)} + \text{True Negatives (TN)} + \text{False Negatives (FN)}}$$

F1-Score

The F1-Score is the harmonic mean of Precision and Recall, providing a balanced measure of the model's performance.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}}$$

3 Rand Index for Clustering Evaluation

3.1 Definition of Rand Index

The Rand Index (RI) is a metric used to evaluate the similarity between two different clusterings of the same dataset. It quantifies the proportion of pairs of data points that are classified consistently across both clusterings. A pair of data points is considered consistently classified if they are either in the same cluster in both clusterings or in different clusters in both clusterings.

Mathematically, the Rand Index is defined as:

$$RI = \frac{a + b}{a + b + c + d} = \frac{\text{number of agreeing pairs}}{\text{total number of pairs}}$$

where:

- *a*: Number of pairs of data points that are in the same cluster in both clusterings (True Positives).
- *b*: Number of pairs of data points that are in different clusters in both clusterings (True Negatives).
- *c*: Number of pairs of data points that are in the same cluster in the first clustering but in different clusters in the second clustering (False Positives).
- *d*: Number of pairs of data points that are in different clusters in the first clustering but in the same cluster in the second clustering (False Negatives).

The value of the Rand Index ranges from 0 to 1, with 1 indicating identical clusterings and 0 indicating no agreement.

3.2 Difference between Rand Index and Accuracy

While both the Rand Index and accuracy are measures of performance, they are applied in distinct machine learning paradigms and evaluate different aspects of model output.

Accuracy is primarily used in **classification** (supervised learning). It measures the proportion of correctly classified individual data points compared to the true labels:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

Accuracy assesses the correctness of assigning individual instances to predefined categories (classes). It requires the knowledge of true labels for each data point.

In contrast, the **Rand Index** is specifically designed for evaluating **clustering** results (unsupervised learning). It assesses the agreement between two ways of grouping data points into clusters, without relying on predefined class labels. The RI focuses on the pairwise consistency of assignments. It evaluates whether pairs of points are grouped together or separated in the same way by both clustering solutions.

The key differences can be summarized as follows:

- **Task Domain:** Accuracy is for classification, while the Rand Index is for clustering.
- **Evaluation Unit:** Accuracy evaluates individual data point classifications, whereas the Rand Index evaluates pairs of data points.

- **Dependence on Ground Truth:** Accuracy requires true class labels for evaluation. The Rand Index can compare two clustering outputs without needing external class labels, although it can also be used with a ground truth clustering if available.
- **Label Sensitivity:** Accuracy is sensitive to the specific labels assigned to classes. The Rand Index is invariant to the actual labels of the clusters; it only considers the groupings.

3.3 Related Discussion

The Rand Index provides a valuable way to quantify the similarity between different clustering outcomes. However, it has a notable limitation: it does not account for the possibility of agreement occurring by chance. Even random clusterings can exhibit a certain level of agreement, leading to a potentially inflated Rand Index value.

To address this, the **Adjusted Rand Index (ARI)** was developed. The ARI corrects the RI for chance agreement, providing a more robust measure of clustering similarity. The ARI ranges from -1 to 1, where 1 indicates perfect agreement, 0 indicates agreement equivalent to chance, and negative values indicate agreement worse than chance.

In situations where the goal is to evaluate the quality of a clustering algorithm against a known ground truth clustering (where the "true" groupings are available), the ARI is often preferred over the raw Rand Index due to its correction for chance. However, the fundamental difference remains that the Rand Index and its variants are tools for assessing the similarity or quality of data groupings (clusters), while accuracy is a measure of the correctness of individual classifications against known categories.

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4.3 Instructive Examples of Rand Index

Let's consider a dataset with 5 data points: $\{p_1, p_2, p_3, p_4, p_5\}$. We have two different clustering results for this dataset:

Clustering 1: $\{\{p_1, p_2\}, \{p_3, p_4, p_5\}\}$ **Clustering 2:** $\{\{p_1\}, \{p_2, p_3\}, \{p_4, p_5\}\}$

We need to consider all possible pairs of data points, which is $\binom{5}{2} = 10$ pairs.

Let's analyze each pair:

1. (p_1, p_2) : Same cluster in Clustering 1, different in Clustering 2 (a increases by 1).
2. (p_1, p_3) : Different in Clustering 1, different in Clustering 2 (b increases by 1).
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Counting the occurrences: $a = 4$, $b = 3$, $c = 0$, $d = 3$.

The Rand Index for these two clusterings is:

$$RI = \frac{a + b}{a + b + c + d} = \frac{4 + 3}{4 + 3 + 0 + 3} = \frac{7}{10} = 0.7$$

This indicates a 70

Example 2: Comparing to a Ground Truth

Let's consider the same 5 data points and assume we have a "true" clustering (ground truth):

Ground Truth Clustering: $\{\{p_1, p_2\}, \{p_3\}, \{p_4, p_5\}\}$ **Clustering Result:** $\{\{p_1\}, \{p_2, p_3\}, \{p_4, p_5\}\}$

Again, we analyze all 10 pairs:

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10. (p_4, p_5) : Same in Ground Truth, same in Result (a increases by 1).

Counting the occurrences: $a = 1$, $b = 5$, $c = 1$, $d = 3$.

The Rand Index in this case is:

$$RI = \frac{a + b}{a + b + c + d} = \frac{1 + 5}{1 + 5 + 1 + 3} = \frac{6}{10} = 0.6$$

This indicates a 60

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Counting the occurrences: $a = 4, b = 3, c = 0, d = 3$.
The Rand Index for these two clusterings is:

$$RI = \frac{a + b}{a + b + c + d} = \frac{4 + 3}{4 + 3 + 0 + 3} = \frac{7}{10} = 0.7$$

This indicates a 70% agreement in the pairwise assignments between the two clusterings.

Example 2: Comparing to a Ground Truth

Let's consider the same 5 data points and assume we have a "true" clustering (ground truth):

Ground Truth Clustering: $\{\{p_1, p_2\}, \{p_3\}, \{p_4, p_5\}\}$ **Clustering Result:** $\{\{p_1\}, \{p_2, p_3\}, \{p_4, p_5\}\}$

Again, we analyze all 10 pairs:

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10. (p_4, p_5) : Same in Ground Truth, same in Result (a increases by 1).

Counting the occurrences: $a = 1, b = 5, c = 1, d = 3$.
The Rand Index in this case is:

$$RI = \frac{a + b}{a + b + c + d} = \frac{1 + 5}{1 + 5 + 1 + 3} = \frac{6}{10} = 0.6$$

This indicates a 60% agreement between the clustering result and the ground truth clustering in terms of pairwise assignments. These examples illustrate how the Rand Index quantifies the similarity between clusterings by considering the consistency of how pairs of data points are grouped.

5.4 Related Discussion

The Rand Index provides a valuable way to quantify the similarity between different clustering outcomes. However, it has a notable limitation: it does not account for the possibility of agreement occurring by chance. Even random clusterings can exhibit a certain level of agreement, leading to a potentially inflated Rand Index value.

To address this, the **Adjusted Rand Index (ARI)** was developed. The ARI corrects the RI for chance agreement, providing a more robust measure of clustering similarity. The ARI ranges from -1 to 1, where 1 indicates perfect agreement, 0 indicates agreement equivalent to chance, and negative values indicate agreement worse than chance.

In situations where the goal is to evaluate the quality of a clustering algorithm against a known ground truth clustering (where the "true" groupings are available), the ARI is often preferred over the raw Rand Index due to its correction for chance. However, the fundamental difference remains that the Rand Index and its variants are tools for assessing the similarity or quality of data groupings (clusters), while accuracy is a measure of the correctness of individual classifications against known categories.

6 Naive Bayes Classification Example: Predicting Computer Purchase

We will use a training dataset to illustrate the Naive Bayes classification algorithm and predict whether a customer will purchase a computer based on their age and student status.

6.1 Training Data

The training data is given in the following table:

Our goal is to predict whether a customer with the following attributes will buy a computer: (Age=Youth, Student=Yes).

6.2 Naive Bayes Approach

We will use Bayes' theorem with the naive independence assumption:

$$P(\text{Buys Computer}|\text{Age}, \text{Student}) \propto P(\text{Age}|\text{Buys Computer}) \times P(\text{Student}|\text{Buys Computer}) \times P(\text{Buys Computer})$$

We need to calculate $P(\text{Buys Computer} = \text{Yes}|\text{Age} = \text{Youth}, \text{Student} = \text{Yes})$ and $P(\text{Buys Computer} = \text{No}|\text{Age} = \text{Youth}, \text{Student} = \text{Yes})$ and choose the class with the higher probability.

Age	Student	Buys Computer
Youth	No	No
Middle-aged	No	Yes
Middle-aged	Yes	Yes
Senior	No	Yes
Senior	Yes	Yes
Senior	No	No
Youth	No	No
Youth	Yes	Yes
Middle-aged	No	Yes
Youth	Yes	Yes
Senior	Yes	No
Youth	No	Yes
Middle-aged	Yes	Yes
Middle-aged	No	Yes

Table 6: Training Data for Computer Purchase Prediction

6.3 Calculating Prior Probabilities

First, let's calculate the prior probabilities of each class:

$$P(\text{Buys Computer} = \text{Yes}) = \frac{\text{Number of 'Yes' instances}}{\text{Total number of instances}} = \frac{9}{14} \approx 0.643$$

$$P(\text{Buys Computer} = \text{No}) = \frac{\text{Number of 'No' instances}}{\text{Total number of instances}} = \frac{5}{14} \approx 0.357$$

6.4 Calculating Likelihood Probabilities

Next, we calculate the likelihood probabilities of each feature value given each class:

For Buys Computer = Yes:

- $P(\text{Age} = \text{Youth} | \text{Buys Computer} = \text{Yes}) = \frac{\text{Number of 'Yes' instances with Age=Youth}}{\text{Number of 'Yes' instances}} = \frac{2}{9} \approx 0.222$
- $P(\text{Student} = \text{Yes} | \text{Buys Computer} = \text{Yes}) = \frac{\text{Number of 'Yes' instances with Student=Yes}}{\text{Number of 'Yes' instances}} = \frac{6}{9} \approx 0.667$

For Buys Computer = No:

- $P(\text{Age} = \text{Youth} | \text{Buys Computer} = \text{No}) = \frac{\text{Number of 'No' instances with Age=Youth}}{\text{Number of 'No' instances}} = \frac{3}{5} = 0.6$
- $P(\text{Student} = \text{Yes} | \text{Buys Computer} = \text{No}) = \frac{\text{Number of 'No' instances with Student=Yes}}{\text{Number of 'No' instances}} = \frac{1}{5} = 0.2$

6.5 Applying Naive Bayes

Now, we calculate the posterior probabilities for the new instance (Age=Youth, Student=Yes):

For Buys Computer = Yes:

$$P(\text{Buys Computer} = \text{Yes} | \text{Age} = \text{Youth}, \text{Student} = \text{Yes}) \propto \\ P(\text{Age} = \text{Youth} | \text{Buys Computer} = \text{Yes}) \times P(\text{Student} = \text{Yes} | \text{Buys Computer} = \text{Yes}) \times P(\text{Buys Computer} = \text{Yes})$$

$$P(\text{Buys Computer} = \text{Yes} | \text{Age} = \text{Youth}, \text{Student} = \text{Yes}) \propto \frac{2}{9} \times \frac{6}{9} \times \frac{9}{14} = \frac{108}{1134} \approx 0.0952$$

For Buys Computer = No:

$$P(\text{Buys Computer} = \text{No} | \text{Age} = \text{Youth}, \text{Student} = \text{Yes}) \propto \\ P(\text{Age} = \text{Youth} | \text{Buys Computer} = \text{No}) \times P(\text{Student} = \text{Yes} | \text{Buys Computer} = \text{No}) \times P(\text{Buys Computer} = \text{No})$$

$$P(\text{Buys Computer} = \text{No} | \text{Age} = \text{Youth}, \text{Student} = \text{Yes}) \propto \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14} = \frac{15}{350} \approx 0.0429$$

Comparing the (proportional) posterior probabilities: $0.0952 > 0.0429$.

6.6 Conclusion

According to Naive Bayes' Classification, the predicted outcome for a customer who is a Youth and a Student is **Yes** (Buys Computer). In this example, we did not encounter zero probabilities, so Laplace smoothing was not strictly necessary for the calculation, but it is a good practice to consider, especially with smaller datasets.

7 Naive Bayes Classification Example for Introductory Class: Predicting Exam Pass/Fail

Consider a dataset of students with two features: 'Hours Studied' (Categorical: Low, Medium, High) and 'Prior Knowledge' (Yes, No), and a class label 'Pass' (Yes, No) indicating whether they passed an exam. We want to predict if a new student will pass based on their study hours and prior knowledge using Naive Bayes.

7.1 Training Data

The training data is as follows:

We want to predict the 'Pass' label for a new student with 'Hours Studied' = High and 'Prior Knowledge' = No.

Hours Studied	Prior Knowledge	Pass
Low	No	No
Medium	Yes	Yes
High	Yes	Yes
Medium	No	No
High	No	Yes
Low	Yes	No
Medium	Yes	Yes
High	Yes	Yes

Table 7: Training Data for Exam Pass/Fail Prediction

7.2 Naive Bayes Approach

We will use Bayes' theorem with the naive independence assumption:

$$P(\text{Buys Computer} | \text{Age, Student}) \propto P(\text{Age} | \text{Buys Computer}) \times P(\text{Student} | \text{Buys Computer}) \times P(\text{Buys Computer})$$

We need to calculate:

$P(\text{Pass} = \text{Yes} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No})$ and $P(\text{Pass} = \text{No} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No})$.

7.3 Calculating Prior Probabilities

$$P(\text{Pass} = \text{Yes}) = \frac{\text{Number of 'Yes' instances}}{\text{Total number of instances}} = \frac{5}{8} = 0.625$$

$$P(\text{Pass} = \text{No}) = \frac{\text{Number of 'No' instances}}{\text{Total number of instances}} = \frac{3}{8} = 0.375$$

7.4 Calculating Likelihood Probabilities

For Pass = Yes:

- $P(\text{Hours Studied} = \text{High} | \text{Pass} = \text{Yes}) = \frac{\text{Number of 'Yes' instances with Hours Studied=High}}{\text{Number of 'Yes' instances}} = \frac{2}{5} = 0.4$
- $P(\text{Prior Knowledge} = \text{No} | \text{Pass} = \text{Yes}) = \frac{\text{Number of 'Yes' instances with Prior Knowledge=No}}{\text{Number of 'Yes' instances}} = \frac{1}{5} = 0.2$

For Pass = No:

- $P(\text{Hours Studied} = \text{High} | \text{Pass} = \text{No}) = \frac{\text{Number of 'No' instances with Hours Studied=High}}{\text{Number of 'No' instances}} = \frac{0}{3} = 0$
- $P(\text{Prior Knowledge} = \text{No} | \text{Pass} = \text{No}) = \frac{\text{Number of 'No' instances with Prior Knowledge=No}}{\text{Number of 'No' instances}} = \frac{2}{3} \approx 0.67$

7.5 Applying Naive Bayes (Without Smoothing First)

For Pass = Yes:

$$P(\text{Pass} = \text{Yes} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No}) \propto 0.4 \times 0.2 \times 0.625 = 0.05$$

For Pass = No:

$$P(\text{Pass} = \text{No} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No}) \propto 0 \times \frac{2}{3} \times 0.375 = 0$$

Without Laplace smoothing, the probability of 'Pass=No' becomes zero because there are no students who failed and studied for 'High' hours in the training data. This might be an oversimplification.

7.6 Applying Naive Bayes with Laplace Smoothing

Let's apply Laplace smoothing (adding 1 to counts and the number of unique values to the denominator).

Number of unique values for 'Hours Studied' = 3 (Low, Medium, High)

Number of unique values for 'Prior Knowledge' = 2 (Yes, No)

For Pass = Yes ($n_{Yes} = 5$):

- $P(\text{Hours Studied} = \text{High} | \text{Pass} = \text{Yes}) = \frac{2+1}{5+3} = \frac{3}{8} = 0.375$
- $P(\text{Prior Knowledge} = \text{No} | \text{Pass} = \text{Yes}) = \frac{1+1}{5+2} = \frac{2}{7} \approx 0.286$

For Pass = No ($n_{No} = 3$):

- $P(\text{Hours Studied} = \text{High} | \text{Pass} = \text{No}) = \frac{0+1}{3+3} = \frac{1}{6} \approx 0.167$
- $P(\text{Prior Knowledge} = \text{No} | \text{Pass} = \text{No}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$

Now, recalculating the posterior probabilities with Laplace smoothing:

For Pass = Yes:

$$P(\text{Pass} = \text{Yes} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No}) \propto 0.375 \times 0.286 \times 0.625 \approx 0.067$$

For Pass = No:

$$P(\text{Pass} = \text{No} | \text{Hours Studied} = \text{High}, \text{Prior Knowledge} = \text{No}) \propto 0.167 \times 0.6 \times 0.375 \approx 0.038$$

Comparing the (proportional) posterior probabilities: $0.067 > 0.038$.

7.7 Conclusion

According to Naive Bayes' Classification with Laplace smoothing, the predicted label for the new student with 'Hours Studied' = High and 'Prior Knowledge' = No is **Yes** (Pass). This example demonstrates a similar level of complexity to the previous one and highlights the importance of handling zero probabilities, which can be a key concept for introductory students to understand.

8 K-Nearest Neighbors (KNN) Algorithm

8.1 KNN Algorithm Pseudocode

Input: Training dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i are feature vectors and y_i are class labels.

Input: A query point q with feature vector x_q .

Input: An integer k , the number of nearest neighbors to consider.

```
1: function KNN(D,  $x_q$ , k)
2:   Initialize an empty list distances.
3:   for each training example  $(x_i, y_i)$  in  $D$  do
4:     Calculate the distance  $d(x_q, x_i)$  between  $x_q$  and  $x_i$  (e.g., Euclidean distance).
5:     Add the tuple  $(d(x_q, x_i), y_i)$  to distances.
6:   end for
7:   Sort the distances list in ascending order based on the distance.
8:   Select the first  $k$  entries from the sorted distances list. These are the  $k$  nearest neighbors.
9:   Initialize a dictionary (or map) class_counts to store the counts of labels among the  $k$  neighbors.
10:  for each of the  $k$  nearest neighbors  $(distance, label)$  do
11:    Increment the count for label in class_counts.
12:  end for
13:  Find the label that has the maximum count in class_counts. This is the predicted class for  $q$ .
14:  return predicted class.
15: end function
```

8.2 Instructive Example: Predicting Sports for Natalie

We want to predict the type of sports for a girl named Natalie, who is 5 years old, using the KNN algorithm with $k = 3$. Gender is encoded as 0 for male and 1 for female. Natalie's features are (Age=5, Gender=1).

8.2.1 Data Table

Here is a data table with 10 rows:

8.2.2 Solving with KNN (k=3)

Natalie's features are $x_q = (5, 1)$. We need to calculate the Euclidean distance between Natalie's features and the features of each person in the training data (Age, Gender).

1. **John (6, 0):** $d = \sqrt{(5-6)^2 + (1-0)^2} = \sqrt{(-1)^2 + 1^2} = \sqrt{1+1} = \sqrt{2} \approx 1.41$ (Football)

Name	Age	Gender	Type of Sports
John	6	0	Football
Alice	5	1	Gymnastics
Peter	7	0	Football
Sophia	4	1	Gymnastics
Michael	6	0	Basketball
Olivia	5	1	Swimming
David	7	0	Football
Emily	4	1	Gymnastics
Daniel	6	0	Basketball
Grace	5	1	Swimming

Table 8: Sports Data

2. **Alice (5, 1):** $d = \sqrt{(5-5)^2 + (1-1)^2} = \sqrt{0^2 + 0^2} = 0$ (Gymnastics)
3. **Peter (7, 0):** $d = \sqrt{(5-7)^2 + (1-0)^2} = \sqrt{(-2)^2 + 1^2} = \sqrt{4+1} = \sqrt{5} \approx 2.24$ (Football)
4. **Sophia (4, 1):** $d = \sqrt{(5-4)^2 + (1-1)^2} = \sqrt{1^2 + 0^2} = 1$ (Gymnastics)
5. **Michael (6, 0):** $d = \sqrt{(5-6)^2 + (1-0)^2} = \sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$ (Basketball)
6. **Olivia (5, 1):** $d = \sqrt{(5-5)^2 + (1-1)^2} = \sqrt{0^2 + 0^2} = 0$ (Swimming)
7. **David (7, 0):** $d = \sqrt{(5-7)^2 + (1-0)^2} = \sqrt{(-2)^2 + 1^2} = \sqrt{5} \approx 2.24$ (Football)
8. **Emily (4, 1):** $d = \sqrt{(5-4)^2 + (1-1)^2} = \sqrt{1^2 + 0^2} = 1$ (Gymnastics)
9. **Daniel (6, 0):** $d = \sqrt{(5-6)^2 + (1-0)^2} = \sqrt{(-1)^2 + 1^2} = \sqrt{2} \approx 1.41$ (Basketball)
10. **Grace (5, 1):** $d = \sqrt{(5-5)^2 + (1-1)^2} = \sqrt{0^2 + 0^2} = 0$ (Swimming)

Now, we sort the distances and select the $k = 3$ nearest neighbors:

1. Alice (5, 1): distance = 0, Sports = Gymnastics
2. Olivia (5, 1): distance = 0, Sports = Swimming
3. Grace (5, 1): distance = 0, Sports = Swimming
4. Sophia (4, 1): distance = 1, Sports = Gymnastics
5. Emily (4, 1): distance = 1, Sports = Gymnastics
6. John (6, 0): distance ≈ 1.41 , Sports = Football

7. Michael (6, 0): distance ≈ 1.41 , Sports = Basketball
8. Daniel (6, 0): distance ≈ 1.41 , Sports = Basketball
9. Peter (7, 0): distance ≈ 2.24 , Sports = Football
10. David (7, 0): distance ≈ 2.24 , Sports = Football

The 3 nearest neighbors and their sports are:

- Alice: Gymnastics
- Olivia: Swimming
- Grace: Swimming

The class labels among the 3 nearest neighbors are: Gymnastics, Swimming, Swimming.

The most frequent class is Swimming (occurs 2 times), followed by Gymnastics (occurs 1 time). Therefore, according to the KNN algorithm with $k = 3$, the predicted type of sports for Natalie is **Swimming**.

8.3 Homework Problem for Introductory ML Class

A university wants to predict whether a student will get admitted to a specific program based on their GPA and standardized test scores. The 'Admitted' field is binary (Yes=1, No=0). Using the K-Nearest Neighbor (KNN) algorithm with $k = 5$, predict the admission status for a new student with GPA = 3.5 and Test Score = 75.

8.3.1 Data Table

Here is a data table of 8 students:

GPA	Test Score	Admitted
3.2	70	0
3.8	85	1
3.0	65	0
3.6	80	1
3.4	72	0
4.0	90	1
3.3	75	0
3.7	82	1

Table 9: Student Admission Data

8.3.2 Hints to Solve

1. The query point for the new student is (GPA=3.5, Test Score=75).
2. Calculate the Euclidean distance between the query point and each student's (GPA, Test Score) in the training data. The formula for Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.
3. Create a list of these distances along with the corresponding 'Admitted' status.
4. Sort this list in ascending order based on the distance.
5. Select the top $k = 5$ entries from the sorted list (the 5 nearest neighbors).
6. Count the number of 'Yes' (1) and 'No' (0) in the 'Admitted' column among these 5 neighbors.
7. The predicted admission status for the new student will be the class (Yes or No) that appears most frequently among the $k = 5$ neighbors. If there is a tie, you can use a tie-breaking rule (e.g., choose the class of the nearest neighbor).

9 Gini Index and Purity in Decision Trees

9.1 Definition of Gini Index

The Gini Index is a measure of the impurity of a dataset or a node in a decision tree. It quantifies the probability of a randomly chosen element being incorrectly classified if it were randomly labeled according to the distribution of labels in the subset. A lower Gini Index indicates higher purity. For a dataset D with m classes, the Gini Index is calculated as:

$$Gini(D) = 1 - \sum_{i=1}^m p_i^2$$

where p_i is the proportion of instances in D that belong to class i .

For a split of a dataset D into two subsets D_{left} and D_{right} with sizes $|D_{left}|$ and $|D_{right}|$ respectively, the Gini Index of the split is the weighted average of the Gini Indices of the two subsets:

$$Gini_{split}(D) = \frac{|D_{left}|}{|D|} Gini(D_{left}) + \frac{|D_{right}|}{|D|} Gini(D_{right})$$

9.2 Purity

Purity refers to the homogeneity of the labels within a subset of data. A subset is considered pure if all its instances belong to the same class. Higher purity generally leads to better classification. Measures like the Gini Index and Entropy

are used to quantify impurity, and thus, lower values of these measures indicate higher purity.

9.3 Problem Solution: Comparing Two Splits

We have a dataset of 100 samples with two classes, A and B. We are given two different ways to split this dataset.

9.3.1 Split 1

Left node (D_{left1}): 60 samples (35 from Class A, 25 from Class B)

- Total samples in $D_{left1} = 60$
- Proportion of Class A in $D_{left1} = p_{A,left1} = \frac{35}{60}$
- Proportion of Class B in $D_{left1} = p_{B,left1} = \frac{25}{60}$
- $Gini(D_{left1}) = 1 - ((\frac{35}{60})^2 + (\frac{25}{60})^2) = 1 - (\frac{1225}{3600} + \frac{625}{3600}) = 1 - \frac{1850}{3600} = 1 - 0.5139 \approx 0.4861$

Right node (D_{right1}): 40 samples (20 from Class A, 20 from Class B)

- Total samples in $D_{right1} = 40$
- Proportion of Class A in $D_{right1} = p_{A,right1} = \frac{20}{40} = 0.5$
- Proportion of Class B in $D_{right1} = p_{B,right1} = \frac{20}{40} = 0.5$
- $Gini(D_{right1}) = 1 - ((0.5)^2 + (0.5)^2) = 1 - (0.25 + 0.25) = 1 - 0.5 = 0.5$

Gini Index for Split 1:

$$Gini_{split1} = \frac{60}{100}Gini(D_{left1}) + \frac{40}{100}Gini(D_{right1}) = 0.6 \times 0.4861 + 0.4 \times 0.5 = 0.29166 + 0.2 = 0.49166$$

9.3.2 Split 2

Left node (D_{left2}): 50 samples (40 from Class A, 10 from Class B)

- Total samples in $D_{left2} = 50$
- Proportion of Class A in $D_{left2} = p_{A,left2} = \frac{40}{50} = 0.8$
- Proportion of Class B in $D_{left2} = p_{B,left2} = \frac{10}{50} = 0.2$
- $Gini(D_{left2}) = 1 - ((0.8)^2 + (0.2)^2) = 1 - (0.64 + 0.04) = 1 - 0.68 = 0.32$

Right node (D_{right2}): 50 samples (15 from Class A, 35 from Class B)

- Total samples in $D_{right2} = 50$
- Proportion of Class A in $D_{right2} = p_{A,right2} = \frac{15}{50} = 0.3$

- Proportion of Class B in $D_{right2} = p_{B,right2} = \frac{35}{50} = 0.7$
- $Gini(D_{right2}) = 1 - ((0.3)^2 + (0.7)^2) = 1 - (0.09 + 0.49) = 1 - 0.58 = 0.42$

Gini Index for Split 2:

$$Gini_{split2} = \frac{50}{100} Gini(D_{left2}) + \frac{50}{100} Gini(D_{right2}) = 0.5 \times 0.32 + 0.5 \times 0.42 = 0.16 + 0.21 = 0.37$$

9.3.3 Determining Better Purity

A lower Gini Index indicates higher purity of the split. Comparing the Gini Indices of the two splits:

$$Gini_{split1} \approx 0.4917$$

$$Gini_{split2} = 0.37$$

Since $Gini_{split2} < Gini_{split1}$, **Split 2** results in better purity.

9.4 Homework Problem: Comparing Information Gain

Consider a dataset of 20 students with a binary target variable 'Interested in AI' (Yes/No) and two features: 'Programming Experience' (Yes/No) and 'Math Skills' (High/Low). The dataset is split based on 'Programming Experience':

Original Dataset (20 students):

- Interested in AI = Yes: 12
- Interested in AI = No: 8

Split based on 'Programming Experience': Left node (Programming Experience = Yes, 10 students):

- Interested in AI = Yes: 9
- Interested in AI = No: 1

Right node (Programming Experience = No, 10 students):

- Interested in AI = Yes: 3
- Interested in AI = No: 7

Calculate the Entropy of the original dataset and the Information Gain of the split based on 'Programming Experience'. Explain how Information Gain relates to the purity of the split. (Note: Entropy of a dataset S with classes $c_1, c_2, \dots, c_m$ is $Entropy(S) = -\sum_{i=1}^m p_i \log_2(p_i)$).